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Mechanical Characterization of Aluminium Alloy 6061 Powder Deposit Made by Friction Stir Based Additive Manufacturing

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Abstract. Additive Friction Stir (AFS) has the potential for extensive future application in metal based additive manufacturing. Powder based AFS is specifically useful for fabricating functionally graded structures. But, the consolidation of powder inside the hollow tool used in this operation hinders the powder based AFS process. This problem could be resolved by Additive Friction Stir Processing (AFSP) while maintaining the key advantages of AFS. A 3D deposit structure of height 5 mm and width 64 mm was made from Al6061 alloy powder by AFSP. Mechanical properties like ultimate tensile strength, yield strength and micro-hardness of the deposit were evaluated in both longitudinal and transverse directions. The ultimate tensile strength and micro-hardness of the deposit were comparable to Al6061-O and there was a significant increment in tensile yield strength. Also, the isotropic nature of the deposit could be inferred from similar mechanical properties in the longitudinal and transverse direction. Dimple ruptures seen in fractographic analysis gave evidence to the ductile nature of the deposit.

Introduction

Additive Friction Stir (AFS) developed by Aeroprobe Corporation is a state-of-the-art friction stir based additive manufacturing technology which is used to negate most of the problems faced during other solid-state additive manufacturing techniques [1] and common fusion based technologies [2]-[4] used in the industries for metal additive manufacturing. In AFS, solid rod or powder filler material is plunged through a hollow tool to the substrate and thereby fabricating the solid deposit with the help of the friction stirring action. Primitive work in this field was done by Kandasamy et al. [5] who have demonstrated the additive manufacturing using AA6061-T6 aluminium alloy and AZ31B and AZ31-H24 magnesium alloy. Tensile strength of the deposited AA6061-T6 was comparable to AA6061-O after solutionizing. At the same time, the tensile strength of AZ31B and AZ31-H24 deposit was close to and more than that of the base material respectively. Also, from the experiments, they have come to a conclusion that the tool geometry and tool traverse rate are the main parameters to control the powder deposition rate. Calvert et al. [6] used WE43 alloys as solid filler rod and powder form in AFS. Deposited materials exhibited isotropic tensile properties. Similar results were observed by Rivera et al. [7] who have used solid filler rod of IN625 on to a substrate of HY80 to perform this friction stirring based additive manufacturing. Use of AFS in the making of Powder Metallurgy aluminium alloys with the addition of transition metals like Ti, Cr, Mn, Mo, Fe, Ni, Si etc. was explored by Daye et al. [8] Mechanical properties became higher due to less thermal processing. The Ultimate Tensile Strength (UTS), Yield Strength (YS), hardness and Young's modulus of the aluminium transition metallurgy powders were more than that of extruded powder.

It can be seen from the literature that most of the works on AFS was done using solid filler rods. Powder based AFS was implemented in case of aluminium with transition metals and magnesium alloys despite its predicted effectiveness in fabricating functionally graded structures. Consolidation of powder inside the AFS tool poses a problem in reaching its full potential. Transfer of heat from the work-tool interface to the AFS tool sintered the powder inside the tool. Hence, the exit of the tool gets choked and the process becomes more complex when removing this consolidation from inside the tool. To maintain the advantages of the AFS process as well as to negate these problems, the authors

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have previously developed a novel technique called Additive Friction Stir Processing (AFSP). In this process, there is a simple modification of pouring the powder from the side of the tool instead of pouring through it. The working principle of AFSP was originated from Friction Stir Processing (FSP). However, the objective of AFSP is to form a 3D deposit rather than modification of the properties of the substrate surface. This process was further modified slightly in this present work by the authors. This modified AFSP was explored with respect to Al6061 alloy powder in order to find out the feasibility of the method.

The objectives of the present work were (i) successful implementation of AFSP in producing deposit layers from Al6061 alloy powder, (ii) mechanical characterization of the additive layers by investigating its mechanical properties such as ultimate tensile strength, yield strength and micro-hardness, and (iii) fractographic analysis to find out the nature of failure.

Experimental Details

Performing AFSP. Consolidation of powder inside the AFS tool was successfully negated by the proposed process.

Aluminium alloy 6061 powder of particle size 50 µm was used as filler material in AFSP. In this present work, AFSP was implemented using a vertical milling machine (Make: Tengzhou Wellon Machinery Co. Model: XL5036B). The schematic and actual photograph of the process along with the final product is shown in Fig. 1.

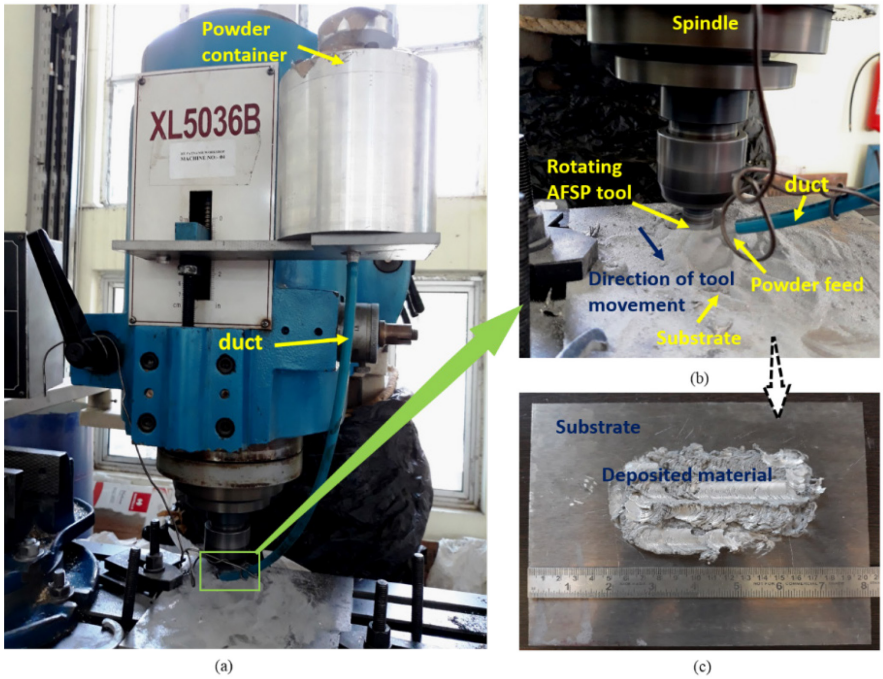

Fig. 1. (a) The actual photograph of the AFSP experimental setup, (b) Close-up view of the working zone and (c) deposit on the substrate

The additive layers were formed on a 10 mm X 200 mm X 200 mm Al6061-T6 substrate. Four tracks of width 16 mm and height 5 mm were formed side by side in order to check the feasibility of the process. The experimental setup shown in Fig. 1 consists of (i) a duct to deliver the powder from the container to the working zone. (ii) a rotating AFSP tool to perform the process. A probe-less cylindrical tool made of H13 steel consisting of a concave bottom and chamfered corner is used in this process. The dimensions of the AFSP tool used is shown in Fig. 2 below.

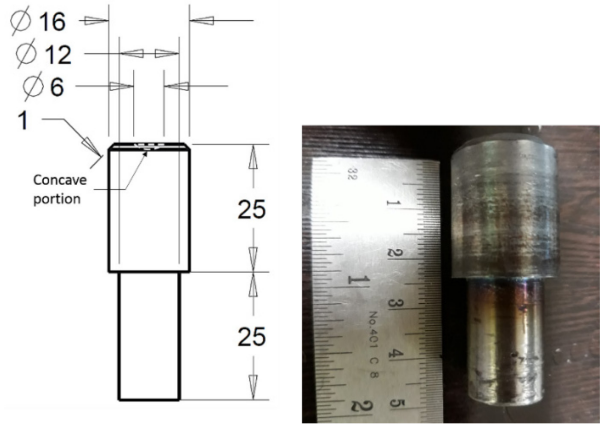

Fig. 2. Schematic of the AFSP tool (All dimensions are in mm) along with actual photograph

Aluminium alloy 6061 powder was filled into the container and made to flow just ahead of the tool by gravitational force through a duct. The tool was rotated over the substrate for some time to pre-heat the surface. This pre-heating resulted in easy plasticization of the powder during the friction stirring operation. Also, it aided in preventing the dispersion of powder particles from the working zone. It was necessary because there was a high tendency of dispersion during the early stages of the operation due to the smoothness of the substrate surface. But, as the initial layers were formed, this dispersion got minimized because the subsequent deposit surfaces were textured. This textured surface helped in holding the powder more firmly. The dispersion was also due to the centrifugal action during the rotation of the tool. The concave bottom surface of the tool helped to prevent this dispersion to some extent. The chamfered corner of the tool gave some easy passage to the filler powder to the working zone which is in between the tool and the substrate.

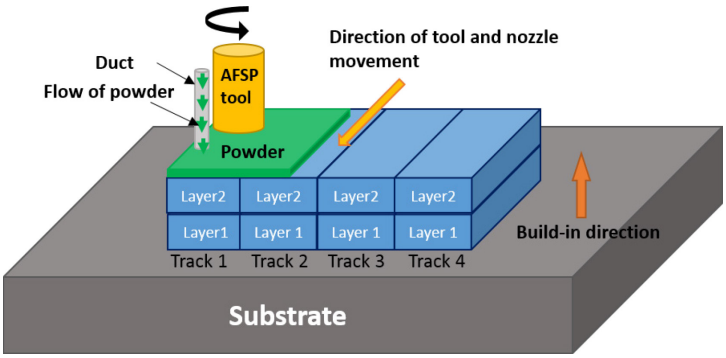

Fig. 3. A demonstration of the AFSP process

A detailed demonstration of the AFSP process is shown in Fig. 3. The thickness of one deposit layer was maintained by a suitable gap between the tool and the substrate surface. This gap was maintained at 0.2 mm for the current experiment. Once, the powder got trapped in this gap, it was plasticized by the stirring operation of the rotating tool and a layer was formed. After a layer was formed, the tool was taken to the side parallel to the previous track and another layer was formed. This operation was repeated 4 times to form 4 initial parallel tracks. After these tracks were formed, the tool was raised by 0.2 mm and taken just above the first layer. The subsequent layer was formed similarly just above this first layer. The powder which was left after the formation of a layer, remained at the top of this layer and was used in the formation of the subsequent layer. The same cyclical procedure was repeated for 25 times to form a 3D structure of height 5 mm, and width 16 mm (same as the diameter of the tool) and also 4 such tracks were formed side by side. The process parameters used for the experiment was tool rpm of 1700 and traverse speed of 22 mm/min. These process parameters were finalized by conducting some sets of trial experiments.

Details of Mechanical Characterization. Tensile specimens and samples needed for micro-hardness testing were cut from the deposit and substrate as shown in Fig. 4.

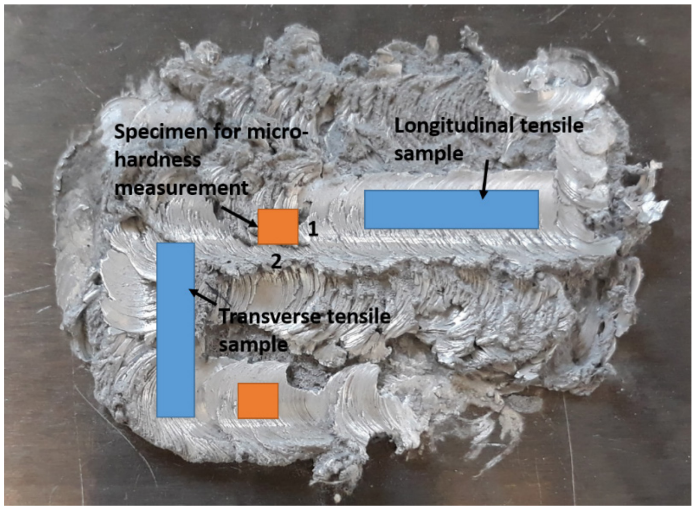

Fig. 4. Demonstration of different mechanical characterization samples cut from the deposit

Ultimate tensile strength and percent elongation of the deposit was found out using a universal tensile testing machine (make: ZWICK/ROELL; model: Z050) using strain rate at 0.001/s. Sub-size tensile specimens were cut in both longitudinal and transverse direction from the deposit. The tensile specimens had a gauge length of 9.5 mm, a width of 3 mm and thickness of 1 mm which was used by Kumara et al. [9]

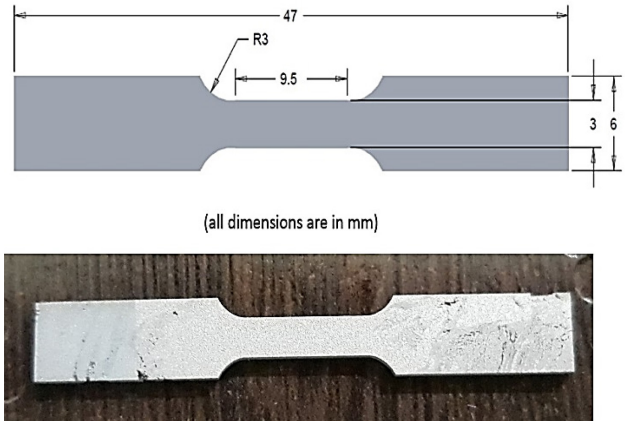

Fig. 5. Schematic and photograph of the tensile test specimen

The tensile specimens were cut from the deposit and workpiece with the help of a wire-cut EDM machine (make: Electronica; model: SPRINTCUT). Sample specimen along with the schematic is shown in Fig. 5.

Samples having the shape of a cuboid with a side of 5 mm and length throughout the total thickness of the deposit and substrate is cut from the specimen. Micro-hardness of the deposit in both longitudinal (1) and transverse (2) directions (refer to Fig. 4) were found out using these samples with the help of a Vicker's micro-hardness testing machine (model: ZHμ/ST2000; make: Zwick/Roell) with 500 gf vertical load. The fractographic analysis of broken tensile specimens was carried out using a Field Emission Scanning Electron Microscope (make: HITACHI; model: S4800).

Results and Discussion

Using AFSP, aluminium 6061 alloy powder was deposited onto a substrate of Al6061-T6 plate of thickness 10 mm, length 200 mm and width 200 mm. After the fabrication of the additive layers, the deposit was subjected to mechanical characterization and fractographic study.

Analysis of Tensile Strength and Yield Strength. The Engineering Stress – Engineering Strain behaviour of the longitudinal and transverse tensile specimens are shown in Fig. 6.

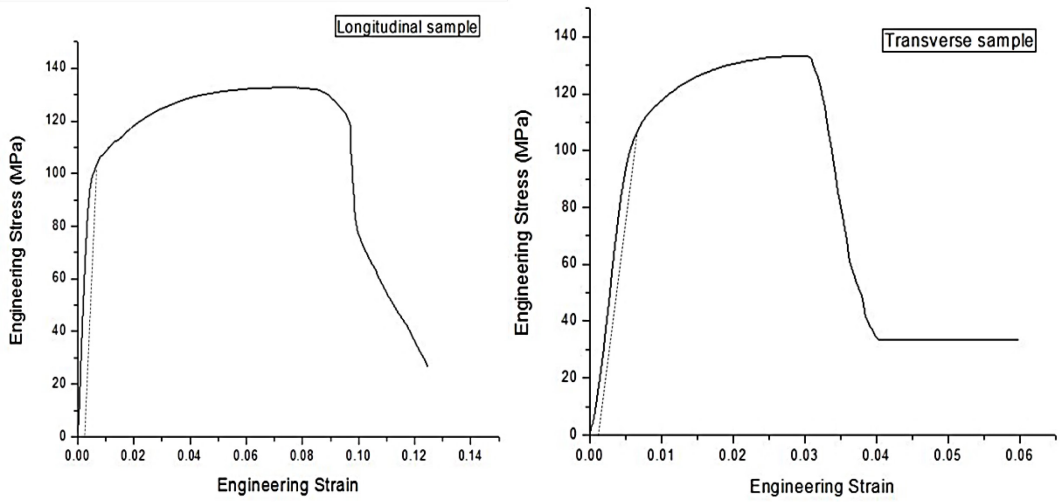

Fig. 6. Tensile behaviour of the deposit in longitudinal and transverse direction

The results of the tensile test were compared with Al6061-O after solutionizing which is given in Table. 1.

Table 1. Comparison of mechanical properties of Al6061-O as per ASTM [10] and the deposit

Average Ultimate Tensile Strength (MPa)			Average Tensile Yield Strength (MPa)		
Al6061-O	Deposit (Longitudinal)	Deposit (Transverse)	Al6061-O	Deposit (Longitudinal)	Deposit (Transverse)
124	132.77	133.35	55.2	102	108

From the table, it is seen that the ultimate tensile strength of the deposit in the longitudinal and transverse direction are 132.77 and 133.35 MPa respectively. These values were comparable to aluminium alloy 6061 after solutionizing. Tensile yield strength was increased by 84.8% and 95.6% respectively in longitudinal and transverse direction. Tensile strength was supposed to become lower due to the thermal effect of the friction stirring. But, the grain refinement and dynamic recrystallization associated with the process led to this increment in strength. It can be said the deposit could be further strengthened by post-deposition aging. The imparted isotropy could be observed from similar values of tensile strengths in the longitudinal and transverse direction. Isotropic properties could be helpful for designing purpose as it makes the fabrication of 3D parts more flexible.

Fractographic Analysis. Ductile failure was observed when fracture surfaces of the tensile specimens were subjected to the Scanning Electron Microscope as shown in Fig. 7. The fractographic analysis showed dimple ruptures both in longitudinal and transverse tensile specimens. This type of failure indicated the ductile nature of the deposit. So, it can be concluded the failure properties of the aluminium 6061 alloy was not changed after deposition.

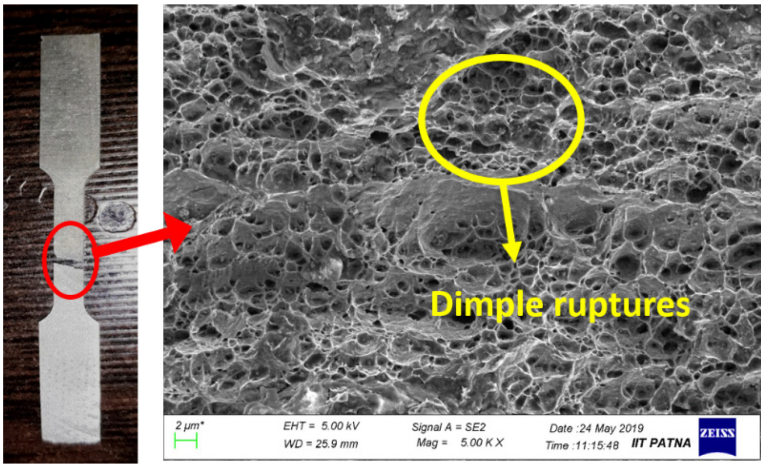

Fig. 7. SEM image of the fracture surface of the tensile specimens cut from deposited layer

Analysis of Micro-hardness. Four readings of Vicker's micro-hardness were taken in the deposit region both in longitudinal and transverse directions and the average of these values were calculated. It was found to be 60.25 HV in the longitudinal direction and 70.25 HV in the transverse direction. From the datasheet, it can be seen that the hardness of Al6061-O is 64 HV.[10] So, it can be concluded that the negative thermal effects on hardness were mostly nullified by imparted grain refinement associated with friction stirring. Due to the grain refinement, the deposited layers are also expected to resist against corrosion.

Conclusions

1. Consolidation of powders hinders the AFS process. This can be successfully negated by AFSP. 5 mm thick deposit structures with 4 parallel tracks were made of aluminium 6061alloy powder by a slightly modified AFSP. A concave geometry at the bottom surface was given to the tool and the substrate surface was pre-heated in order to reduce the dispersion of powder from the working zone.
2. The average ultimate tensile strength of the deposit was slightly more than that of solutionized aluminium 6061 alloy both in longitudinal and transverse directions. The values in both directions were similar which proved the isotropic nature of the deposit. Also, the average tensile yield strength of the deposit exceeded those of Al6061-O by 84.8 and 95.6% in longitudinal and transverse directions.
3. The average Vicker's micro-hardness in the longitudinal and transverse direction of the deposit was found to be 60.25 and 70.25 HV respectively whereas the value for Al6061-O is 64 HV.
4. Fractographic analysis proved that the failure of the deposit was ductile in nature.
5. Functionally graded structures could be manufactured by the AFSP technique.

Acknowledgments

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